



جامعة الأمير محمد بن فهد
PRINCE MOHAMMAD BIN FAHD UNIVERSITY

Cafeteria Management System

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Individual contributions of each team member

Arwa I Alqassab: Supplement Interface and Request Management:

- I created a while loop so the admin and user can navigate between options by using the switch statement, and I did the menu for the admin and the user so the admin first will have to choose the first one, which is Drop down menu for the Admin and the second one is Go back to Main Menu if the admin clicks on the first option, it will go to the admin menu, and it has a. Add new food and its information, b. Delete the information about the food, c. Search for a particular food variety, d. Check the number of food varieties, e. Generate reports of food, and f. Go back to the menu also if the admin clicks (a) will have
(1. Add new food, 2. Display pending requests, and 3. Add pending requests to the menu)

For Administrators:

- I created a method that sees all pending requests (displayPendingRequests) I created this method by using a linked list and by using while loop and an if statement
- I created a method (choosePendingRequestToAddToMenu) so the admin can choose the Request that want to add to the menu. By plugging this feature, the admin takes a made list and loads food kinds that are from a variety directly to the menu (displayPendingRequests)
- By deciding about this option, directors can centre on a list of pending applications, upload authorized food sorts to a menu, and prevent any discrepancies with the order.
- The administrators will assign each detail (code, title, origin, calories, and price)
- Manage pending requests: The managers can analyze and match the already queued request line having a different range of food on the menu.
- Display pending requests: This preference comes up with a provision listing the details of all the pending requests for further administrative reviews.
- Remove pending requests: A decision maker can lead off the grid depending on signals from waiting applicants.
- Check modern-day food sorts: Administrators can simply reassure themselves on whether they offer various meals or not by simply checking the menus in the menu. I did it by using linked list, a while loop and an if statement

For User:

- After having the user visit the device, it provides you with a variety of choices.
- The menu affords two alternatives: We have "Drop down menu for User" and "Go once again to Main", choosing "Drop down menu for User" will lead the user to the option of interacting with various users and doing different activities.
- In the "Drop down menu for User" menu, the user has the following options: (a. add Request, b. Add Complaint, and c. Go back to menu)
- I created "requestNewFood" This method, on one hand, allows user to go into the variety of food choices I use the scanner to do it and by use method to check if the food on the menu or not if the food is on menu the user will see a message like "The requested food variety already exists in the menu." If the food is not on the menu the user will see messages like "Request for new food variety submitted successfully."

Here, when a customer puts up a requirement, it is directly moved to the listing of pending requests. Next, administrators can apply this "choosePendingRequestToAddToMenu"

Fager A Alyati:

- I wrote two listnode methods. deletefood method and addfood method in the administration class.
- In the addFood method, I use the insertfront method. This method is responsible for adding a new food item to the menu. It takes six parameters: code, title, origin, variety, calories, and price, representing the details of the new food item. Inside this method it creates a new food object then it creates a new listnode holding the food object. First, it will check if the head of the linked list is null. If it is null it means the list is empty it will set the head to the newnode. After updating the linked list it will open the file food_details.txt, write the details of the new food item and will print added successfully finally it will close the file and catch the IOException or InputMismatchException.
- In the deletefood method, I use the find method and delete method together to make it a delete at position method but instead of position, I used the code number. In this method, it will delete food item from the menu based on only one parameter "foodCode". Inside this method, it converts foodCode to lowercase for case-insensitive comparison. Then using the while loop it will traverse through the linked list to find the matching code. If the code item is found it will remove it from the listnode and update the linkedlist then it will open the file food_details.txt then it will update the file with the remaining food items then it will catch any IOException finally it will print deleted successfully whether the food item was found and deleted or if it wasn't found in the menu.

Syeda F. Mohiuddin:

- Developed the project timeline & ensured the project progressed accordingly (using Trello)
- Defined class **food** with data members (*code, title, origin, variety, calories, & price*)
- declared as *private*, and a parameterized constructor to initialize them
- Provided **setter & getter methods** to assign and return the data members (class *food*)
- Provided **toString()** method to return the string representation of food object (class *food*)
- Defined class **listnode** to represent a node in the linked list, which stores food object *data* and *link* to the next node, and a parameterized constructor to initialize them
- Implemented **searchVariety(String food_var)** in class *Administrator* that searches for a specific variety type
- Implemented **displayTotalVariety()** in class *Administrator* that calculates & displays the total number of different food varieties
- Implemented **generateReport(String choice)** & helper methods (for easy generation of reports on other choices/data members integrated in future) - **repTitle(), repOrg(),**

repVar(), repCal(), repPri() - in class *Administrator* that displays data based on food's title, origin, variety, calories, and price

- Implemented **emptyFile()** in class *Administrator* to empty all contents of *food_details.txt* file when exiting the application to remove the entry of old food details
- Reviewed and finalized the project (.java file & report) before submitting

Zainab Fatima Wajid:

Complains:

- Created a case 'c' in a switch that belongs to the *displayUserMenu()* method which functions to take the user's complaints.
- First, it asks the user for the '**Food Title**' and if the food title is in the list it asks the user for "**Food item Complain title**" and takes the input from the user. After the user is done entering the complaint it displays "**Complaint Added Successfully.**"
- Secondly, if the food title is not in the list the system displays "**Sorry, the food title is not available. Please check your spelling or inquire about the food first.**"

Method displayFoodDetails():

- The displayFoodDetails() method reads food data from a file, prints each food item's code and title to the support, and handles any file reading errors.
- It opens the file, reads each line, splits it into code and title, and then prints both. It finally closes the file reader.
- If an issue occurs when reading a file, an error message is displayed.

Class User:

- Created inquireFood method with parameters input and admin, in the class User.
- This method asks the user to provide the code for the requested food item.
- It moves through a linked list of food items stored in the Administrator class.
- At each stage, it compares the entered food code to the current food item's code.
- When it finds a match, it shows the specifics of the recognized food item and sets the discovered value to true.
- If no matching item is identified it displays "**The food is not in the menu.**"